Detecting Exoplanets with the Keeble Telescope

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Abstract

This project discusses the results of attempting to detect exoplanets with the Keeble telescope and demonstrates the methods involved. An exoplanet, or a planet that exists outside of the solar system, is typically detected through the transit method. With the transit method, a CCD can image an exoplanet's host star during the exoplanet's expected transit time. Once calibrated, the reduced science images undergo photometry, and the pixel values for the host star and its neighboring comparison stars are read in to create a light curve. This light curve is direct evidence that an exoplanet has been detected. Therefore, the project's basis is to explore the possibility of finding exoplanets by using the transit method with the equipment in the Keeble observatory and analyzing the data collected to derive light curves. Many improvements have been made to heighten the chances of fulfilling this goal, such as automating the dome to aid in telescope tracking and collimating for refining the imaging quality and/or focus. Fortunately, there has been success in finding definitive transiting exoplanets and several light curves have been produced from the data obtained.

1 Introduction

In 1992, the first planet outside the solar system was discovered; since then, hundreds of these distant planets, called "exoplanets", have been discovered, observed, and cataloged (Cowley and Hughes). The most popular and effective method for detection is through the "transit method", which is an eclipse-like phenomenon where the object temporarily blocks the light emitted by its host star. As the observer on Earth, scientists can use ground-based telescopes to record a change in flux: a fraction as small and sensitive as 0.25 (Bettex). The transit method can determine many characteristics of an exoplanet, including its size, speed of orbit, and even the presence of an atmosphere. Aside from the physical

traits of the exoplanet, the data of these recurring dips in brightness can be recorded as graphs known as light curves, a product of photometry (NASA). Specifically, differential photometry will compare the light curves of a host star to its neighboring stars (or "comp" stars, for comparison) during an imaging session (Conti). Viewing a transiting exoplanet can be done by space telescopes like the James Webb Space Telescope (JWST) and the Transiting Exoplanet Survey Satellite (TESS) and like previously mentioned, a ground-based telescope too.

Currently, TESS is traveling through space, searching for exoplanets to survey with the transit method and photometry. Thousands have already been cataloged, and likely thousands of more will be; these numbers are merely stepping stones in the goal of 200,000 candidate stars (Garner). Although the level of space-based technology is quite advanced, ground-based telescopes play an important role in compiling exoplanet catalogs. Ground-based telescopes can provide support with follow-up observations to confirm the findings of TESS. The partnership of TESS and ground-based telescopes will not only lead to the culmination of new exoplanetary discoveries, but will also characterize the exoplanets by determining geological and/or atmospheric compositions.

As a ground-based telescope, Keeble observatory has the potential to aid TESS. Despite just recently celebrating its 60th anniversary, Keeble observatory was rebuilt in 2017. This means the new observatory is updated with technology that features more advanced optics and reliable machinery. Certainly it could be assumed that with its updated features, the Keeble telescope would be able to join TESS and/or similar exoplanet groups. However, there is an obstacle: the Keeble telescope has never attempted to find an exoplanet transit, so it is uncertain whether it can. The main objective, before joining a group to do follow-up observations, is to figure out if Keeble can confidently make appropriate observations in the first place. Furthermore, this project investigates and confirms the possibility of exoplanet detection with the Keeble telescope.

2 Data

Over the span of nine weeks, 10 observations were attempted (*see Table 1*).

Observing Run	Completed (Yes / No)	Date	Exposure time (Seconds)	Exposure temperature (Celsius)
TOI-1259-Ab	Yes	6/9-6/10	600	-9°
WASP-14b	Yes	6/10-6/11	180	-9°
HAT-P-37b	Yes	6/13-6/14	180	-9°
TrES-3b	Yes	6/14-6/15	180	-9°

TOI-1259-Ab	No	6/16-6/17	180	-9°
KELT-21b	Yes	6/28-6/29	180	-7°
TOI-4463Ab	No	7/6-7/7	120	-5°
WASP-74b	Yes	7/10-7/11	120	-7°
TrES-1b	Yes	7/11-7/12	30	-7°
KELT-16b	Yes	7/12-7/13	30	-5°

Table 1: Observation runs and details.

The table contains details on the completion status, date, exposure time, and exposure temperature for each data run. As the observations continued on, more adjustments were made to the procedure and/or instruments to improve the accuracy and quality of the data. The earlier observations have stars that are unclear and sparse in numbers, which makes it difficult to plate-solve or place apertures for astrometry purposes. Whereas, newer observations have an image quality that is clearer, allowing visibility of a larger amount of more "precise" stars. A recurring issue that is prevalent in both cases is faulty, or even sometimes loss of, telescope tracking. Bad tracking will be further investigated in future research.

To improve image quality, the images are routinely calibrated using flatfields and darkfields. Flatfields are images taken in a limited time frame, during the hours of dawn and dusk. It is important to note that the sky must be illuminated by the sun's light, but not enough to saturate the flatfields. A dark, night sky has stars that will interfere with the calibration process. The ideal flatfield is a blank piece of sky that highlights the imperfections (dust rings, residue or defects on the CCD window) that appear in each image. Clouds and stars will impede the reduced picture's quality. On the other hand, darkfields can only be taken in a completely dark environment. It is similar to the flatfield in the sense its ideal appearance is to be as uniform and blank as possible; but instead, it captures nothingness. The combination of these two types of images are essential for calibration, a process that will be explained in-depth in the methods section.

Once a raw science image undergoes calibration, the reduced data becomes "smooth". All of the blemishes disrupting the original image, such as dust rings, hot pixels, overall unevenness, etc., are removed. The finalized image benefits from calibration since imperfections would hinder the photometry process by interfering with the pixels on an individual basis. Photometry completely relies on

reading values on a pixel-by-pixel basis, so a “clean” image is optimal for accurate results.

3 Methods

3.1 Instruments

The two equipment necessary for this project are the telescope and the CCD. The telescope, located in Keeble observatory, is an ASA 40 cm Ritchey-Chretien F/8. Its movements are controlled by the programs Autoslew and Cartes du Ciel, whereas the shutter and focal adjustments rely on ACC. As for the dome, it operates with ACE SmartDome and ASCOM POTH hub for opening/closing commands and movements.

The CCD, also known as the “Charge-Coupling Device”, is a camera that produces images depending on the amount of light visible in the field of view. As the light is passing into the telescope, photons are captured by a grid of pixels and electrons are generated and freed. The amount of electrons generated are determined by two factors: how much light is passing in and the sensitivity of each pixel. Not all pixels are equal in sensitivity; therefore, images can be susceptible to hot pixels. The presence of hot pixels can hinder the photometry process, so data calibration (specifically darkfields) is essential for the correction of these blemishes. Testing pixel sensitivity is a separate goal from the main objective, because the CCD’s reaction to light on an individual pixel level is unknown.

A SBIG ST-10XMEI is the CCD used for this project. It has a thermoelectric cooler and a pixel resolution of 0.88 arcsec per pixel. The camera functions with Maxim DL for general commands and each image can be viewed and edited in SAOImage ds9. For this project, the lens primarily used was an infrared lens; reasoning behind this is to minimize the effects of light pollution, since only infrared light is able to pass through.

3.2 Observation Procedure

A night of observation starts right before sunset, since that is the optimal time to take flatfields. Although there is the option to collect flatfields before sunrise instead of sunset, it is preferable to take them at both times as a precaution (in the case of weather conditions or bad quality) and comparison reasons. Once the night flatfields are obtained, the telescope is synced to a bright star near the target exoplanet star. Then, the right ascension and declination coordinates for the transiting star are entered into Cartes du Ciel and the telescope is commanded to locate the given position. Once found, the CCD takes exposures to confirm the target star is within the field of view. If correct, then the telescope is nudged to center the object. If not, an exposure is uploaded to Astrometry.net to be assigned

corresponding WCS (World Coordinate System) positions and then, the telescope is moved appropriately.

The now centered transiting star is prepared for exposures. Exposures run for an hour, regardless of the duration for each individual frame. It is structured this way for two reasons: to accommodate for poor tracking and to do routine, hourly check-ups on the telescope. If the target star has moved significantly since the beginning of the observation run, then the frame is moved to prevent it and neighboring, potential comparison stars. The telescope itself is examined at least once an hour to ensure the power cords are not tugging on the camera, since this can meddle with the telescope's motor abilities or pull the CCD off of the telescope. When the telescope hits its limit and flips, this can put the CCD equipment in jeopardy.

Lastly, after a few hours of observation, the images will become too saturated due to the rising sun. Flatfields are taken a second time before sunrise. Dark Fields can be taken on nights of observations as well, but a majority were collected independent of these days. In order to take a darkfield, the surrounding environment must be completely dark and the cap remains on the CCD window opening. These can be done at any time.

3.3 Reduction

As previously mentioned, reduction (also referred to as calibration) benefits data by eliminating imperfections that can hinder photometry results. AstroImageJ is the program responsible for carrying out reduction and additionally, plate-solving. Specifically, the CCD Data Processor function is put into application. Before reducing the image, the data set runs through the plate-solving feature and are given their WCS coordinates. These WCS-solved images are pipelined out into a separate folder than the untampered set of data and then is used for the reduction process. First step of the reduction process is to build master frames for the darkfields that correlate to the flatfields, the flatfields, and the darkfields that correlate to the raw science image.

The first masterframe to be built belongs to the darkfield-flatfield data set. After the correct directory is established and all the files are read in, a singular image is created. Every individual pixel in the master frame has the median exclusively calculated from its affiliated pixels. All the compiled images in the set contribute to the median pixel values on a pixel-by-pixel basis. The freshly created frame is exported to a directory of choosing after completion.

Subsequently, the darkfield-flatfield master frame is used to normalize the flatfields. In order to proceed, it is necessary to check to confirm that the master dark and the read-in flatfields match in exposure time and temperature. This is crucial for the next step of subtraction. Again, subtraction is done on an individual

pixel level; to avoid redundancy, keep in mind that all of the steps in reduction are done through this method. From each flatfield file, the median master frame dark is subtracted, resulting in a sequence of normalized flatfield images. To continue, a selected region (typically the clearest in the frame) is scaled and solved for its median. The original stack of flat fields is divided by the respective normalized files and furthermore, finishing off with combining the files into a masterframe.

Exactly as the first median dark frame is built, the second one is created in a similar fashion. The only exception is that instead of using dark files with an exposure time and temperature resembling the flat fields, it matches the science images. The creation and accessibility of the median master frame darks-science and master flats means all the steps for reduction preparation have been completed. All of the plate-solved science images are read-in and undergo dark-subtraction and flat-division, eliminating the visible defects and smoothing out the raw data. Between the flat field's capability of erasing general lens/window defects and dark field's hot pixel removal, it should be noted that there is one interference that cannot be eliminated: cosmic rays. Cosmic rays remain after calibration and may even render an image unusable. Nonetheless, this is an unlikely occurrence, so the reduction process is still overall beneficial. The finalized images are exported to a folder, ready for photometry.

3.4 Photometry

All of the reduced images are imported back into AstroImageJ, but this time for photometry. The files are read-in as a sequence, organized in numerical order. Each frame can be flipped through and viewed in a slideshow-like set-up. This is perfect for deleting inadequate images that fall victim to clouded weather, loss of good tracking, or other various disturbances prior to alignment. Since all of the images are plate-solved, the sequence can be aligned by using WCS information. After, the now aligned images are saved to a new folder.

Placing apertures on an image can prove to be difficult considering a slight adjustment can change the results– for better or for worse. The first step in the approach is to confirm which star is the target star. The target star is the basis for the light curve, regardless of whether it produces one. Next is to assemble the supporting team: the comparison stars. Getting the "right" comparison stars is vital and there are a multitude of suggested guidelines to exclude ones that may cause faulty results. Variable stars, saturated stars, stars that leave the frame of view, and stars that exceed the linearity limit typically are not suitable.

Apertures work by processing a star's pixel values with an inner aperture and sampling the background values with an outer aperture. While the inner aperture sums the pixel values, the outer aperture takes the individual pixels and divides by median of each value. The photometry performed in this experiment

used a fixed photometric aperture radius of 14 pixels, a fixed inner background annulus of 20 pixels, and a fixed outer background annulus of 30 pixels. These are placed on each of the stars that will be included in the light curve. The amount of comparison stars used for each data set varied, but usually fell in the range of 4 to 6 stars.

Pixels are read-in and a light curve is produced; however, there are still adjustments made. Some tools that prove to be useful for finalizing the light curve are the *Fit and Normalization Region Section, Orbital Parameters, and Host Star Parameters*. The Fit and Normalization Region Section presents the option of trimming the data using visible lines. On each light curve, the left/right trim are placed at start and finish of the data set while the left/right lines highlight the curve 10 minutes before and after the transit. Plotting features aside, the Orbital Parameters and Host Star Parameters aided with producing the experimental results, such as the depth, transit center, and planetary radius. These numbers, produced by the data, are compared to the expected values.

3.5 Improvements

Several modifications were made to the observatory. The first change was to the dome; it was limited to manual controls but is now automated. This was achieved by entering information about the dome and telescope dimensions into the ASCOM POTH hub settings. Another example of an improvement is covering the telescope with a shroud to prevent scattered light from traveling in and interfering with the exposures. This is especially useful while taking flats, since it is not entirely dark at this time in the evening. Regarding the CCD, the camera window has been cleaned to reduce dust rings in images. One of the most noticeable improvements was collimating the telescope. In early observation runs, many exposures contained stars that appear lop-sided or elongated (*see Figure 1*). Although bad tracking has largely contributed to this issue, collimation was a partial solution to the bigger problem at hand: telescope precision. By de-focusing the optics, the primary mirror can be adjusted accordingly to ensure incoming light is as circular and uniform as possible. The images post-collimation have better clarity and more stars visible (*see Figure 2*).

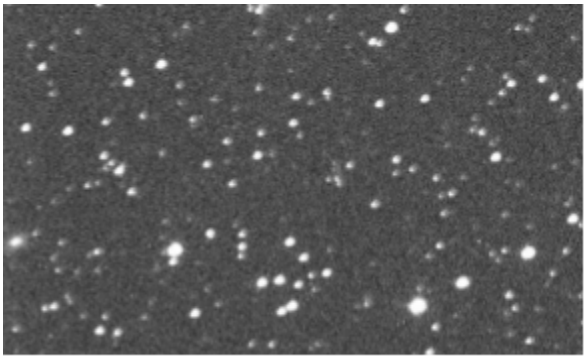


Figure 1: HAT-P-37 b, before collimation (6/13-6/14/23), 180 second exposure.

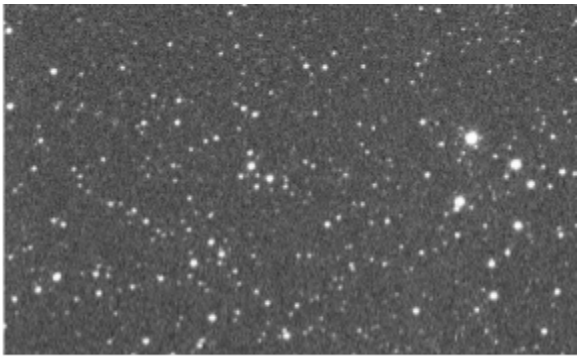


Figure 2: KELT-16 b, after collimation (7/12-6/13/23), 30 second exposure.

4 Results

4.1 TOI-1259 A b

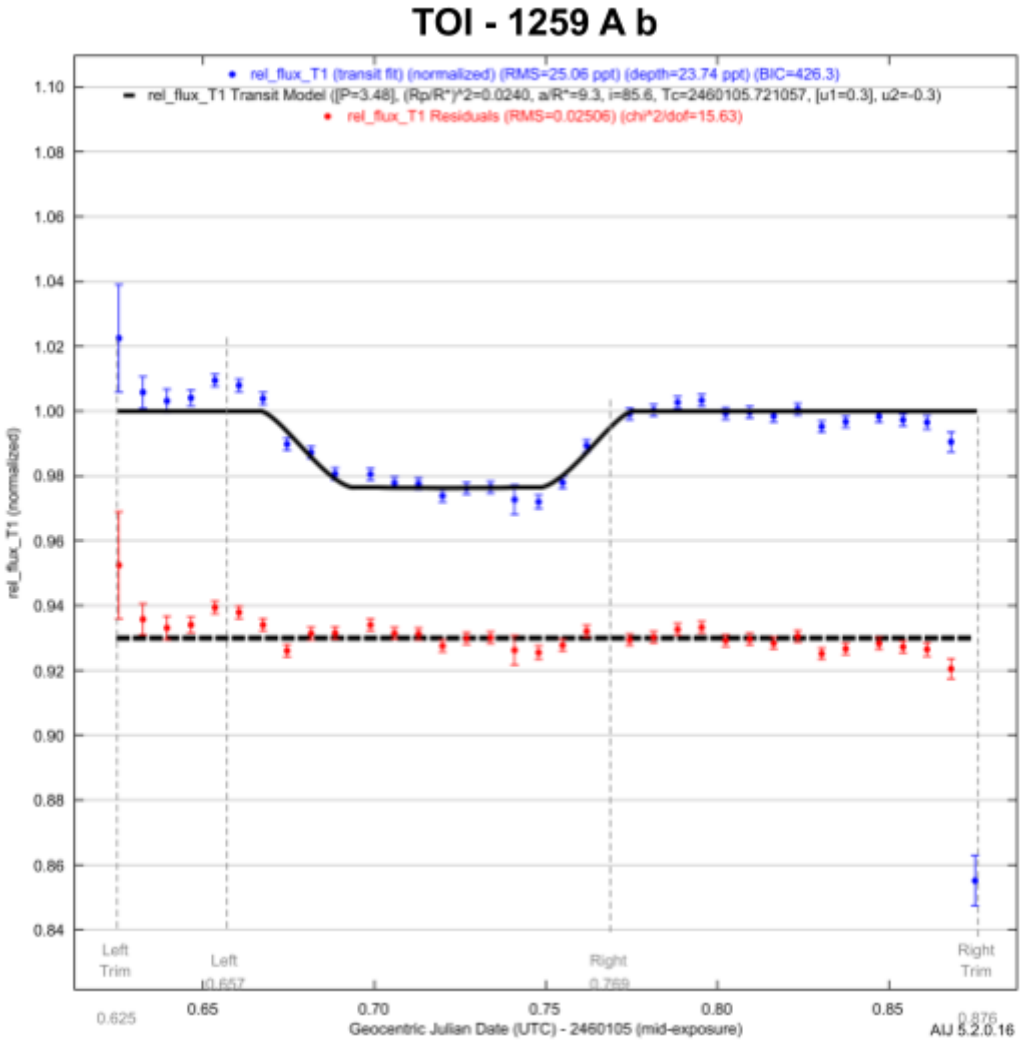


Figure 3: TOI-1259 A b light curve.

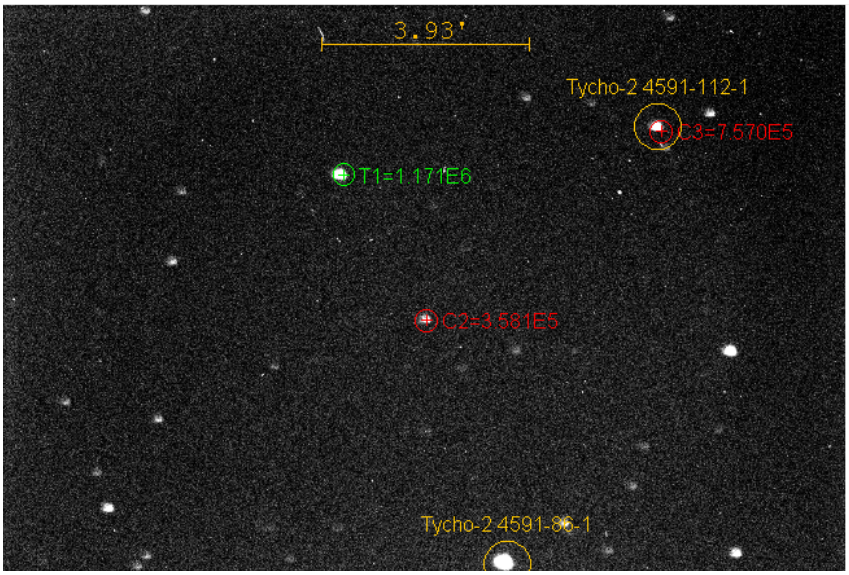


Figure 4: TOI-1259 A b (T1) and comparison stars (C2 to C3)

Host Star		Planet	
Magnitude	12.1	Period	3.48 days
Distance from Earth	118.11 pc	Semi-major Axis	0.407
Temperature	4775 K	Mass	0.44 M_{Jup}
Mass	0.74 $M_{\odot}$	Radius (Expected)	1.02 R_{Jup}
Radius	0.71 $R_{\odot}$	Radius (Experimental)	1.07 R_{Jup}

Table 3: TOI-1259 A b Stellar and Planetary Parameters.

Transit			
Date	6/9-6/10	Depth (Expected)	27 ppt
Duration	2:31 hrs	Depth (Experimental)	23.74
Duration of observation run	10.5 hrs	T_C (UTC) (Expected)	5:12
Transit time (Eastern Time)	23:56--01:12--02:27	T_C (UTC) (Experimental)	5:18

Table 4: TOI-1259 A b transit information.

Science Images		Flatfields	
Exposure time	600 seconds	Exposure time (Morning)	10 second(s)
Temperature	-9°C	Exposure time (Night)	1, 2 second(s)
No. of images	36	Temperature (Morning)	-9°C
Images Deleted	Image 36	Temperature (Night)	-9°C

Table 5: TOI-1259 A b exposure information.

TOI-1259 A b is the first exoplanet attempted to be detected by the Keeble telescope. It is a data set of 36 images and most of the exposures are of low quality from clouded weather conditions and bad tracking. Alongside the tracking issues, the observation log reports that the automated dome wasn't aligned with the telescope and was somewhat off-center. The stars are noticeably elongated and sparse in the exposures. Little to no comparison stars are available. The last two

images were saturated; while one was deleted, the other was kept in the light curve.

A light curve is produced from the data, albeit uncertain how accurate it is due to circumstances of the data. As for the given experimental values, these are decently consistent with the expected numbers. The expected depth is 27 ppt, while the experimental depth is 23.74 ppt. Regarding the planetary radius, expected length is 1.02 R_{Jup} and the calculated experimental is 1.02 R_{Jup}.There is a 6 minute difference in transit center, since the expected time is 5:12 UTC and experimental is 5:18 UTC.

4.2 WASP-14b

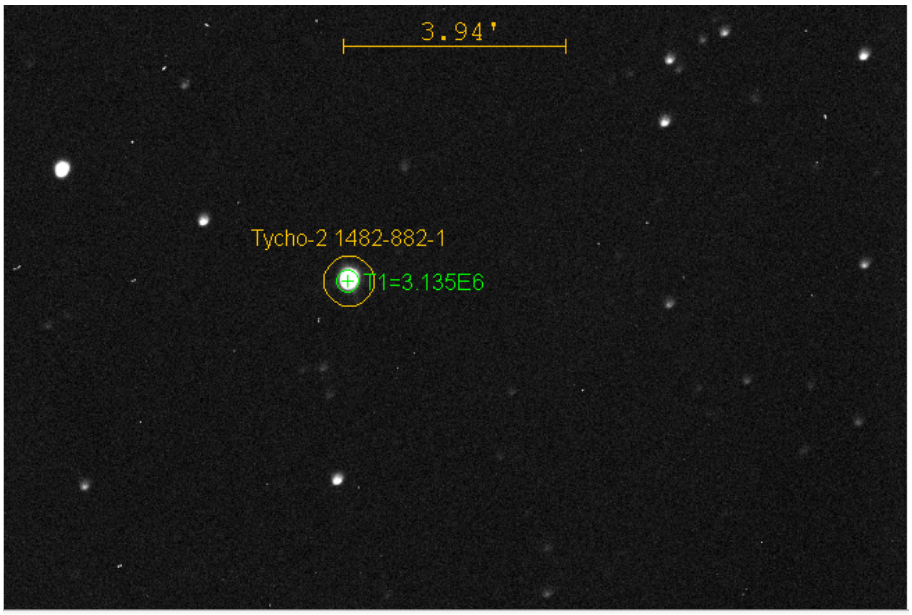


Figure 5: WASP-14b (T1) and comparison stars (N/A)

Host Star		Planet	
Magnitude	9.7	Period	2.24 days
Distance from Earth	162 pc	Semi-major Axis	0.0358

Temperature	6458 K	Mass	8.840±1.400 M_{Jup}
Mass	1.31 $M_{\odot}$	Radius (Expected)	1.38±0.08 R_{Jup}
Radius	1.33 $R_{\odot}$	Radius (Experimental)	N/A

Table 6: WASP-14b Stellar and Planetary Parameters.

Transit			
Date	6/10-6/11	Depth (Expected)	10 ppt
Duration	2:34 hrs	Depth (Experimental)	N/A
Duration of observation run	10.5 hrs	T_C (UTC) (Expected)	3:29
Transit time (Eastern Time)	22:12--23:29--00:46	T_C (UTC) (Experimental)	N/A

Table 7: WASP-14b transit information.

Science Images		Flatfields	
Exposure time	180 seconds	Exposure time (Morning)	N/A
Temperature	-9°C	Exposure time (Night)	1,2 second(s)

No. of images	85	Temperature (Morning)	N/A
Images Deleted	N/A	Temperature (Night)	-6°C

Table 8: WASP-14b exposure information.

WASP-14b is the second attempt for detection. Similarly to the last observing run, a large portion of the exposures are blurred from bad tracking and are sparse in stars. During this transit, the telescope hit its west limit and flipped in response, so tracking was temporarily lost. The dome closed up without being commanded to, resulting in power brick falling. Cartes du Ciel needed the time to be readjusted.

No light curve can be created from the data's current condition. Due to the flip midway through the transit, apertures are unable to be placed, and therefore cannot be aligned or undergo photometry. The data cannot be plate-solved because the stellar positions are unrecognizable to the algorithm since there are not enough stars in frame. This means that alongside the apertures, WCS is unable to align the data set as well.

4.3 HAT-P-37b

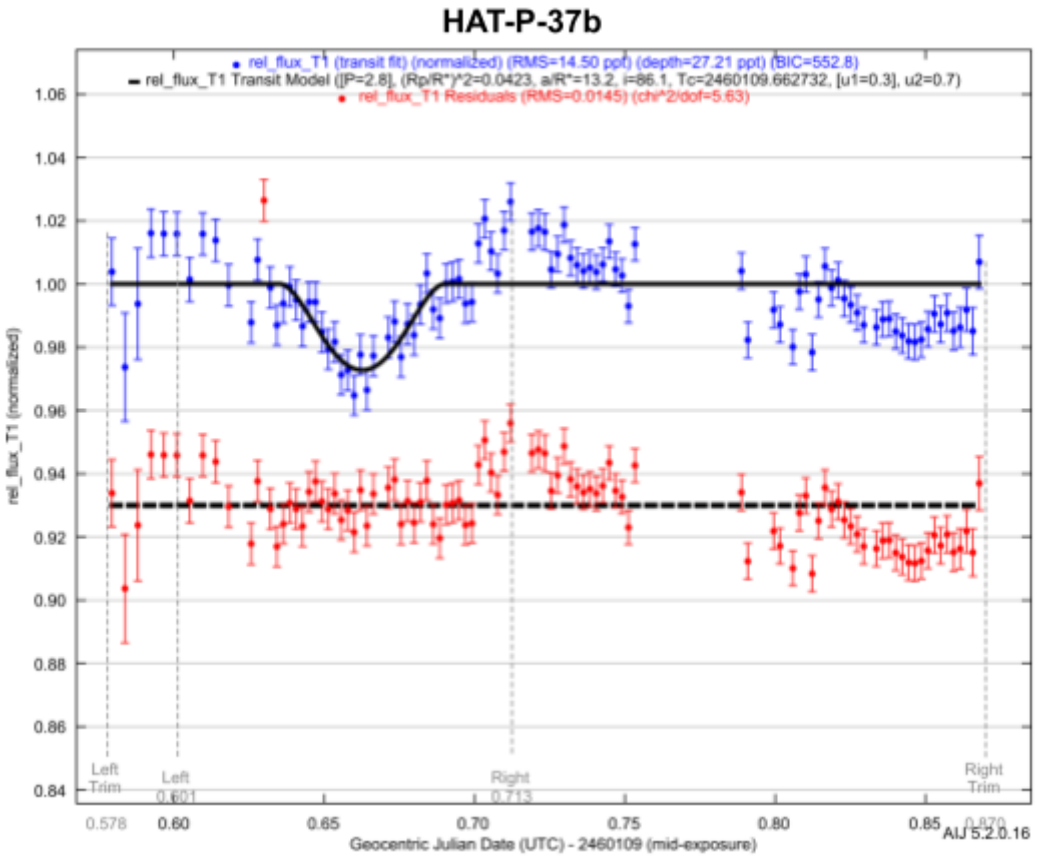


Figure 6: HAT-P-37 b light curve.

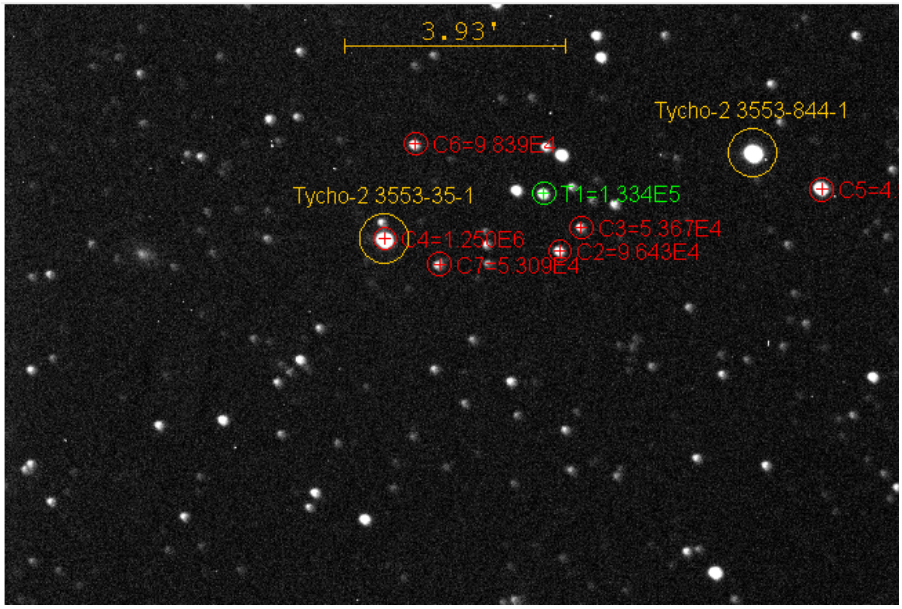


Figure 7: HAT-P-37b (T1) and comparison stars (C2 to C6)

Host Star		Planet	
Magnitude	13.4	Period	2.8 days
Distance from Earth	390.59 pc	Semi-major Axis	0.0379
Temperature	5477 K	Mass	1.169±0.103 R_{Jup}
Mass	0.96 $M_{\odot}$	Radius (Expected)	1.178±0.077 R_{Jup}
Radius	0.87 $R_{\odot}$	Radius (Experimental)	1.73 R_{Jup}

Table 9: HAT-P-37b Stellar and Planetary Parameters.

Transit			
Date	6/13-6/14	Depth (Expected)	19 ppt
Duration	2:20 hrs	Depth (Experimental)	27.21 ppt
Duration of observation run	10.5 hrs	T_C (UTC) (Expected)	3:46

Transit time (Eastern Time)	22:36--23:46--00:56	T_C (UTC) (Experimental)	3:54

Table 10: HAT-P-37b transit information.

Science Images		Flatfields	
Exposure time	180 seconds	Exposure time (Morning)	1,2 second(s)
Temperature	-9°C	Exposure time (Night)	10 second(s)
No. of images	109	Temperature (Morning)	-9°C
Images Deleted	Images 68-72	Temperature (Night)	-7°C

Table 11: HAT-P-37b exposure information.

HAT-P-37b is the third attempt for detection. In this observation run, the exposures are shorter and consequently, the number of images collected has increased compared to the previous two attempts. More stars are visible in this set. Images 68-72 are deemed inadequate due to the blur of the stars, caused by the tracking issues. The observation log recorded that the tracking issues were resolved by relocating and centering the target star back into the frame of view.

A light curve is generated from the available data. As shown, there is a prominent gap between the plotted points around the time when the tracking was lost. The calculated values are as follows .The expected depth is 19 ppt and the experimental data finds a depth of 27.21 ppt. Another comparison is the expected transit center time of 3:46 UTC versus the data's time of 3:54 UTC. Lastly, the data gave a planetary radius measurement of 1.73 R_{Jup}, whereas the expected value is 1.178±0.077 R_{Jup}.

4.4 TrES-3b

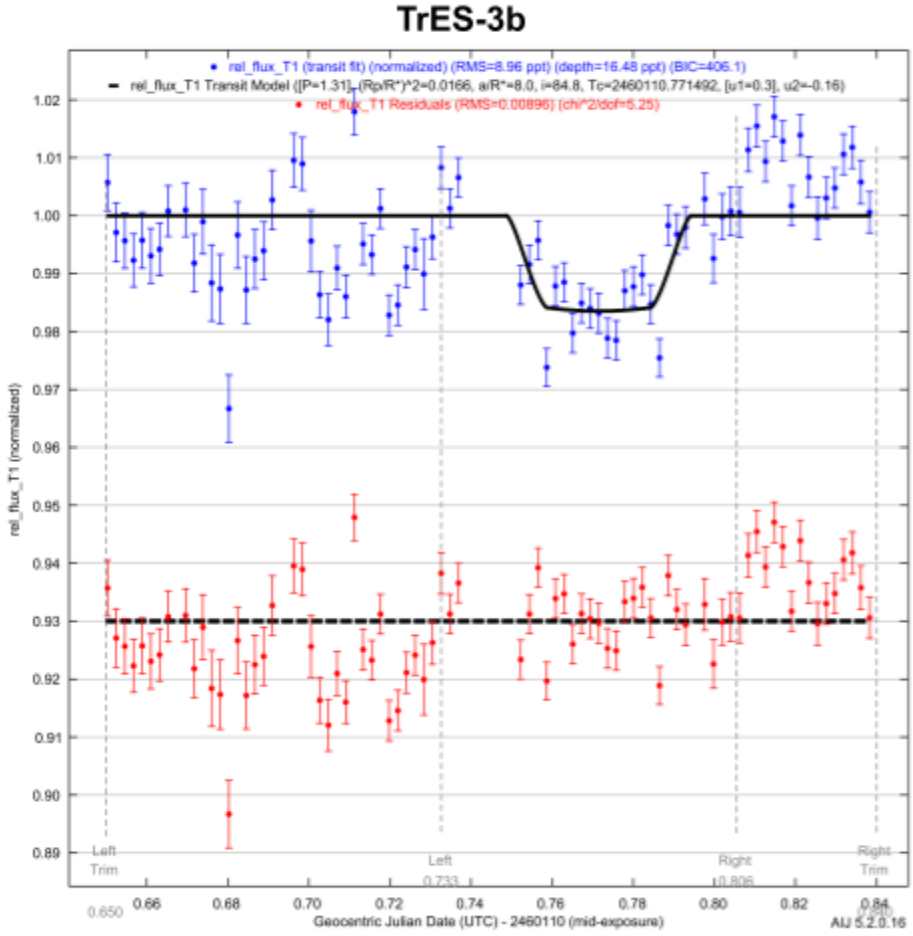


Figure 8: TrES-3b light curve.

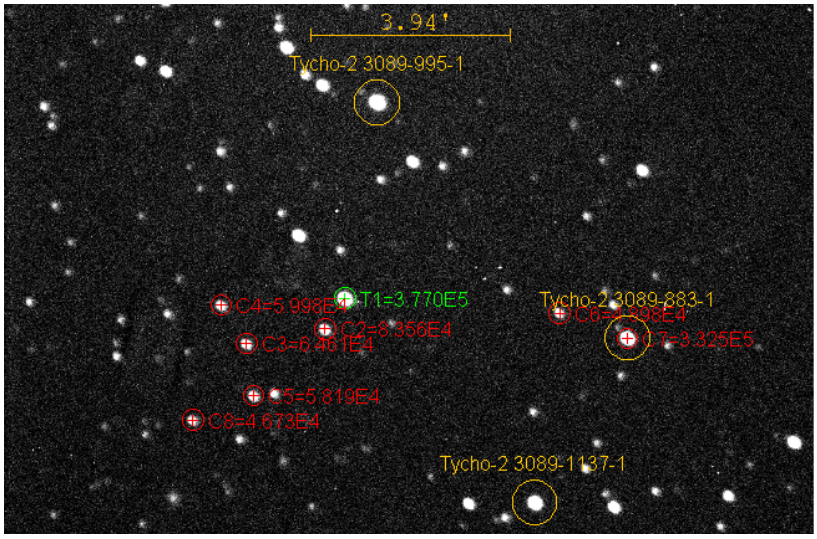


Figure 9: TrES-3b (T1) and comparison stars (C2 to C7)

Host Star		Planet	
Magnitude	12.4	Period	1.31 days
Distance from Earth	pc	Semi-major Axis	0.02282
Temperature	5489 K	Mass	1.91 M_{Jup}
Mass	0.96 $M_{\odot}$	Radius (Expected)	1.34 R_{Jup}
Radius	0.82 $R_{\odot}$	Radius (Experimental)	1.03 R_{Jup}

Table 12: TrES-3b Stellar and Planetary Parameters.

Transit			
Date	6/14-6/15	Depth (Expected)	27.4 ppt
Duration	1:25 hrs	Depth	16.48

		(Experimental)	
Duration of observation run	1:25 hrs	T_C (UTC) (Expected)	6:28
Transit time (Eastern Time)	01:45--02:28--03:10	T_C (UTC) (Experimental)	6:31

Table 13: TrES-3b transit information.

Science Images		Flatfields	
Exposure time	180 seconds	Exposure time (Morning)	N/A
Temperature	-9°C	Exposure time (Night)	2,10 second(s)
No. of images	80	Temperature (Morning)	N/A
Images Deleted	Image 9	Temperature (Night)	-7°C

Table 14: TrES-3b exposure information.

The fourth attempt of detection is TrES-3b. It is a set of 80 images. The exposures appear in a similar manner as HAT-P-37b: more visible stars, although not entirely clear. Image 9 was discarded since the WCS algorithm was unable to recognize the stars in the image. The observation log reports that no notable issues were observed.

A light curve is derived from the available data. The expected depth is 27.4 ppt, while the experimental depth is 16.48 ppt. Regarding the planetary radius, expected length is 1.34 R_{Jup} and the experimental calculated 1.03 R_{Jup}.There is a 3

minute difference in transit center, therefore the expected time of 6:28 UTC and the experimental of 6:31 UTC are comparable.

4.5 TOI-1259 A b

Follow up 6/16-6/17

This is a follow-up observation of TOI-1259 A b. It was unsuccessful for two reasons. Firstly, observations began too late and the transit peak was missed. The second mishap was the dome closing up on itself again. There is no light curve for this observation.

4.6 KELT-21b

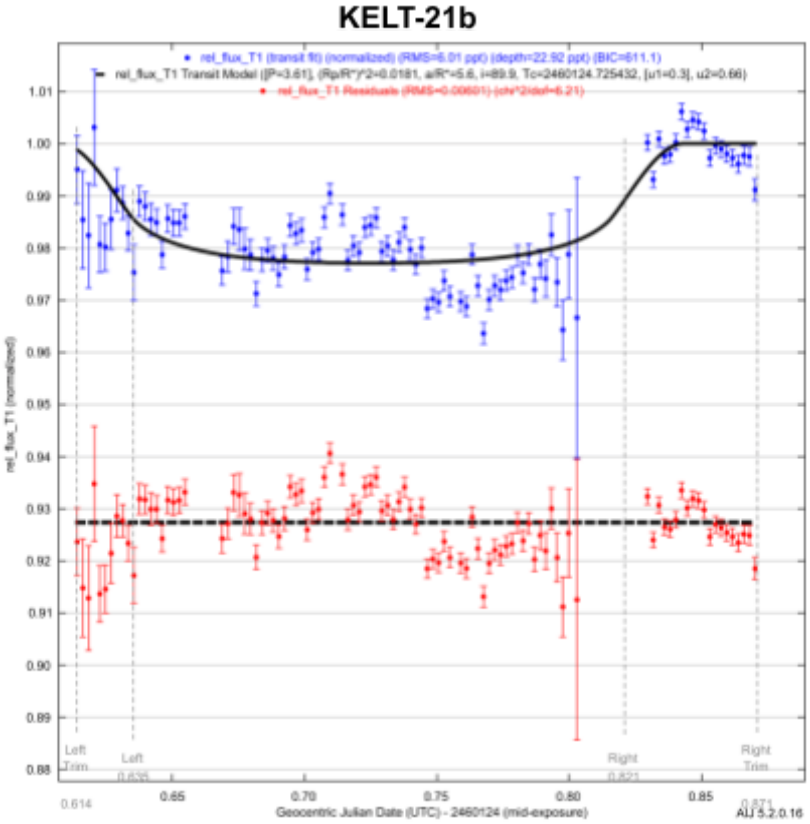


Figure 10: KELT-21b light curve.

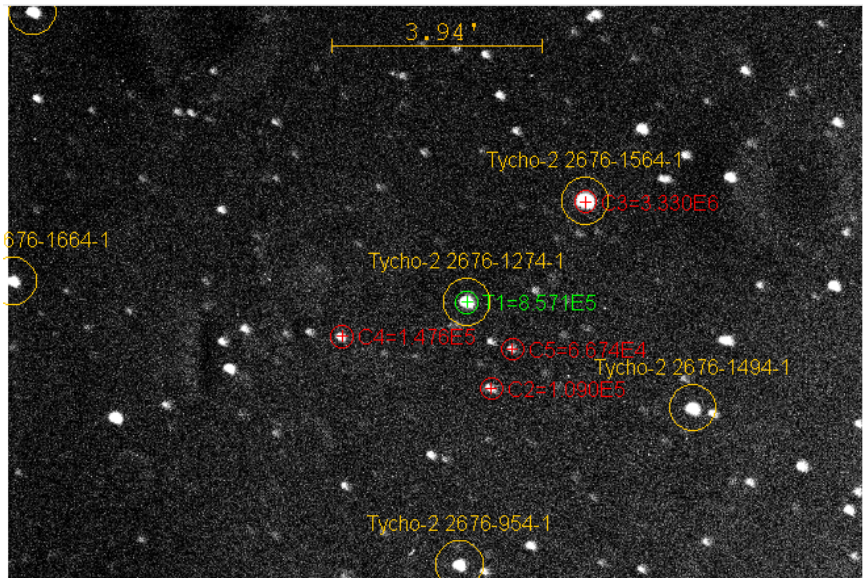


Figure 11: KELT-21b (T1) and comparison stars (C2 to C5)

Host Star		Planet	
Magnitude	10.5	Period	3.61 days
Distance from Earth	470.88 pc	Semi-major Axis	0.05224
Temperature	8324 K	Mass	<3.91 M_{Jup}
Mass	2.05 $M_{\odot}$	Radius (Expected)	1.59 R_{Jup}
Radius	1.7 $R_{\odot}$	Radius (Experimental)	2.24 R_{Jup}

Table 15: KELT-21b Stellar and Planetary Parameters.

Transit			
Date	6/28-6/29	Depth (Expected)	9.9 ppt

Duration	4:07 hrs	Depth (Experimental)	22.92 ppt
Duration of observation run	10.5 hrs	T_C (UTC) (Expected)	5:29
Transit time (Eastern Time)	23:25--01:29--03:32	T_C (UTC) (Experimental)	5:24

Table 16: KELT-21b transit information.

Science Images		Flatfields	
Exposure time	180 seconds	Exposure time (Morning)	1 second
Temperature	-7°C	Exposure time (Night)	1,2 second(s)
No. of images	101	Temperature (Morning)	-7°C
Images Deleted	None	Temperature (Night)	-3,-4°C

Table 17: KELT-21b exposure information.

KELT-21b is the sixth attempt to detect an exoplanet transit. It is significantly longer than the other observation runs at a duration of 4 hours. Morning flats are saturated, so the night flats are used for reduction. Tracking was lost between the hours of 3 and 3:30 EST. A laptop update occurred at 3:15 EST.

There is a light curve for this data set. Some comparisons between expected and experimental values are the following. The data provided a transit depth of 22.92 ppt, deeper than the expected 9.9 ppt. As for the transit center, expected is 05:29 UTC, comparable to the calculated of 05:24 UTC. AstroImageJ gives a planetary radius value of 2.24 R_{Jup} , while the expected measurement is 1.59 $R_{Jup.}$

4.7 TOI-4463 A b

TOI-4463 A b is another failed attempt. The observation run ran into issues with the weather conditions. The target star lost visibility for the peak through the latter half of the transit when clouds rolled in. There is no light curve for this data set.

4.8 WASP-74b

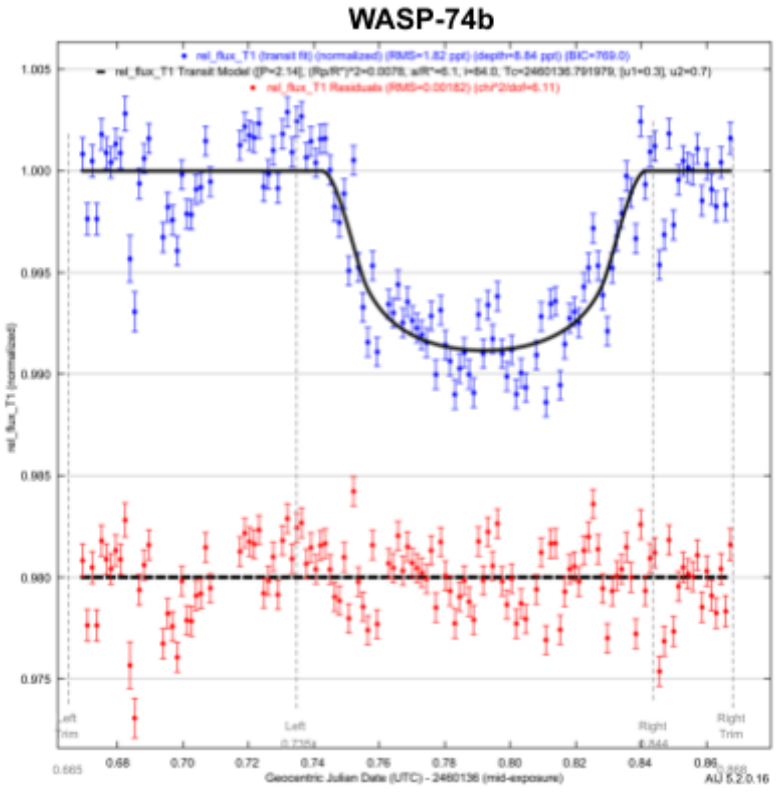


Figure 12: WASP-74b light curve.

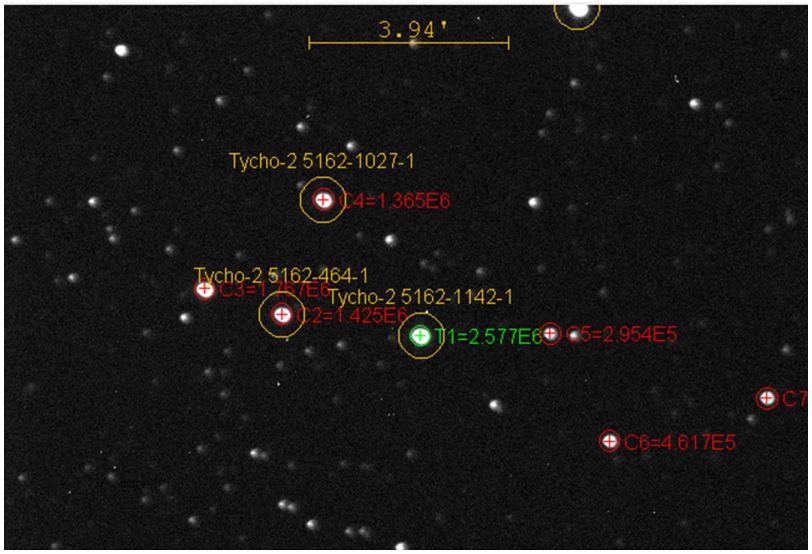


Figure 13: WASP-74b (T1) and comparison stars (C2 to C6)

Host Star		Planet	
Magnitude	9.8	Period	2.14 days
Distance from Earth	149.22 pc	Semi-major Axis	0.03443±0.00036
Temperature	5806 K	Mass	0.72±0.12 M_{Jup}
Mass	1.04 $M_{\odot}$	Radius (Expected)	1.36±0.10 R_{Jup}
Radius	1.6 $R_{\odot}$	Radius (Experimental)	1.3 R_{Jup}

Table 18: WASP-74b Stellar and Planetary Parameters.

Transit			
Date	7/10-7/11	Depth (Expected)	9.6 ppt

Duration	2:17 hrs	Depth (Experimental)	8.84 ppt
Duration of observation run	10.5 hrs	T_C (UTC) (Expected)	6:56
Transit time (Eastern Time)	2:17	T_C (UTC) (Experimental)	7:00

Table 19: WASP-74b transit information.

Science Images		Flatfields	
Exposure time	120 seconds	Exposure time (Morning)	1 second(s)
Temperature	-7°C	Exposure time (Night)	1 second(s)
No. of images	132	Temperature (Morning)	-2°C
Images Deleted	Image 1,2,18,19	Temperature (Night)	-7°C

Table 20: WASP-74b exposure information.

WASP-74b is the eighth attempt for exoplanet detection. Images 1, 2, 18, and 19 have been deleted since these exposures are disturbed by tracking problems. No issues were reported during the observation run.

As shown, the data is sufficient to produce a light curve. The expected depth is 9.6 ppt and the experimental data finds a depth of 8.84 ppt. Another valid comparison is the expected time of the transit center versus the time given by the data. The expected time is 06:56 UTC, meanwhile the calculated peak is 07:00 UTC. Last, but not least, the data gives a planetary radius measurement of 1.3 R_{Jup}, whereas the expected value is 1.36±0.10 R_{Jup}.

4.9 TrES-1b

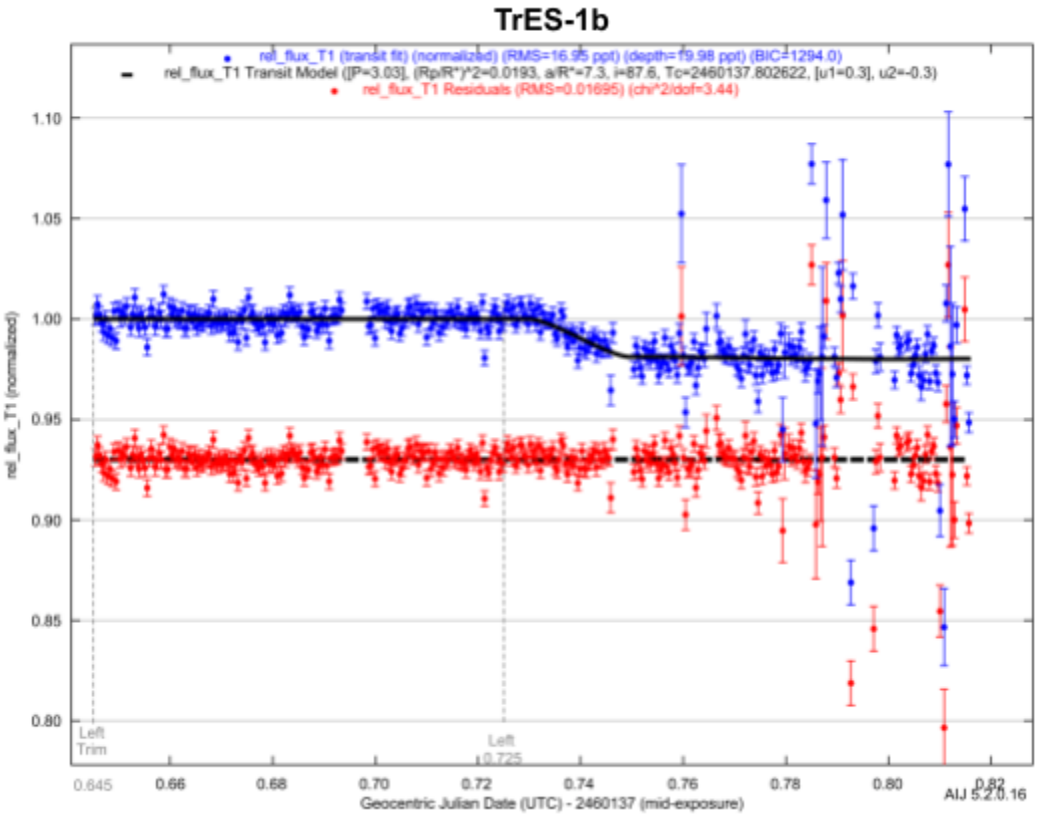


Figure 14: TrES-1b light curve.

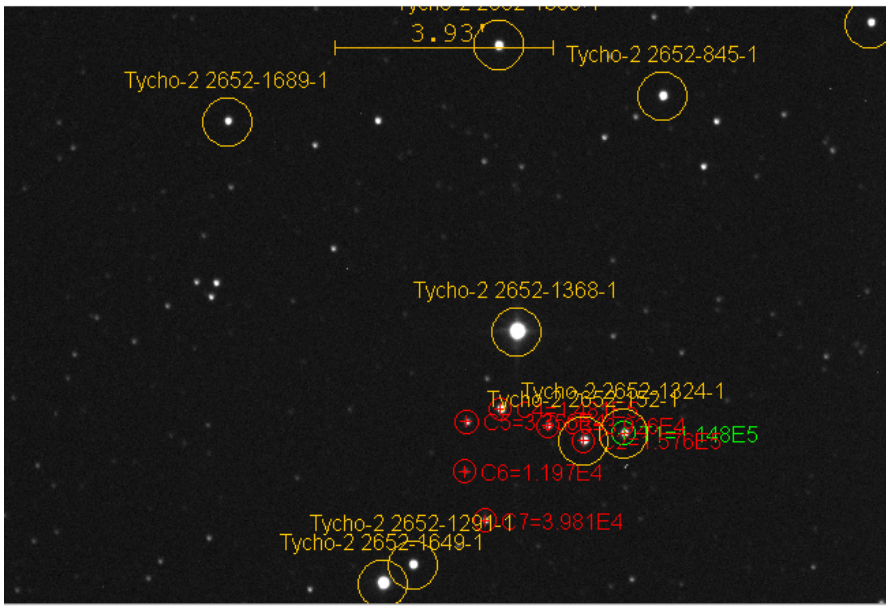


Figure 15: TrES-1b (T1) and comparison stars (C2 to C7)

Host Star		Planet	
Magnitude	11.4	Period	3.03 days
Distance from Earth	159.66 pc	Semi-major Axis	0.03926
Temperature	5242.82 K	Mass	$0.84 \pm 0.11\ M_{Jup}$
Mass	$0.9\ M_{\odot}$	Radius (Expected)	$1.13 \pm 0.06\ R_{Jup}$
Radius	$0.86\ R_{\odot}$	Radius (Experimental)	$1.16\ R_{Jup}$

Table 21: TrES-1b Stellar and Planetary Parameters.

Transit			
Date	7/11-7/12	Depth (Expected)	18 ppt
Duration	2:31 hrs	Depth (Experimental)	19.98 ppt
Duration of observation run	11 hrs	T_C (UTC) (Expected)	6:49
Transit time (Eastern Time)	01:34--02:49--04:05	T_C (UTC) (Experimental)	7:16

Table 22: TrES-1b transit information.

Science Images		Flatfields	
Exposure time	30 seconds	Exposure time (Morning)	1 second(s)
Temperature	-7°C	Exposure time (Night)	1 second(s)
No. of images	505	Temperature (Morning)	-3°C
Images Deleted	Image 17, 277-280,315,328 ,335-337,343-34 5, 365-366, 397-end	Temperature (Night)	-7°C

Table 23: TrES-1b exposure information

TrES-1b is the ninth attempt to detect exoplanet transits. This is the first night of observation post-collimation. Many images were discarded because of the interference from poor weather conditions and tracking issues. Tracking was lost near the end of the transit. All exposures from after the tracking recovery were not reliable data and were excluded. The CCD was moved, therefore camera orientation was not preserved. Thus, the flats from the morning are faulty.

A light curve is produced from the data. Furthermore the given experimental values are comparable with the expected numbers. The expected depth is 18 ppt, while the experimental depth is 19.98 ppt. Regarding the planetary radius, expected length is $1.13 \pm 0.06\ R_{Jup}$ and the calculated experimental is $1.16\ R_{Jup}$.There is a 17 minute difference in transit center, since the expected time is 06:49 UTC and experimental is 07:16 UTC.

4.10 KELT-16b

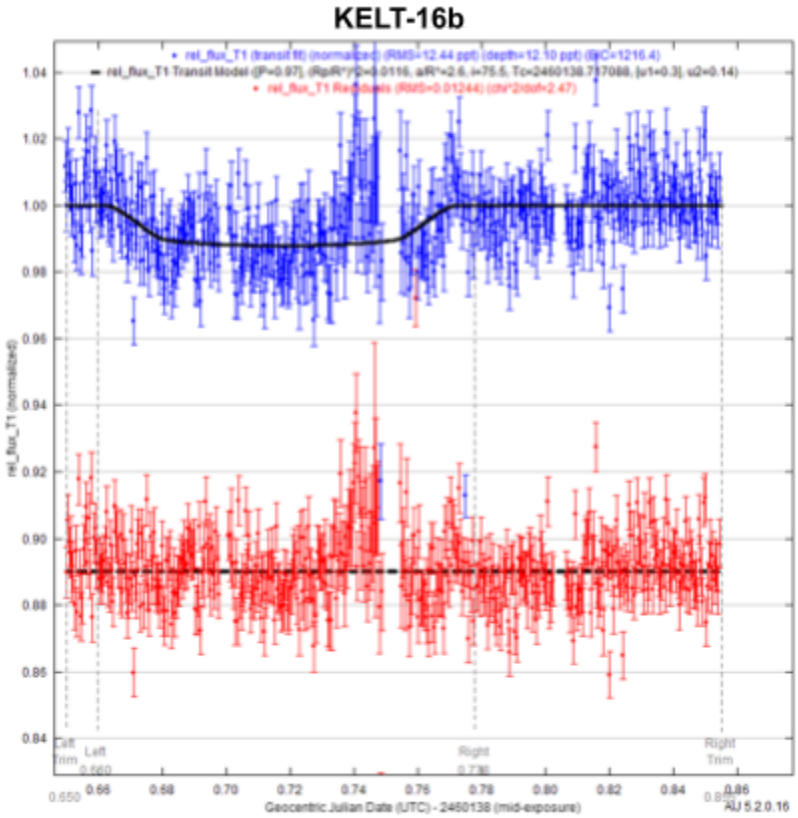


Figure 16: KELT-16b light curve.

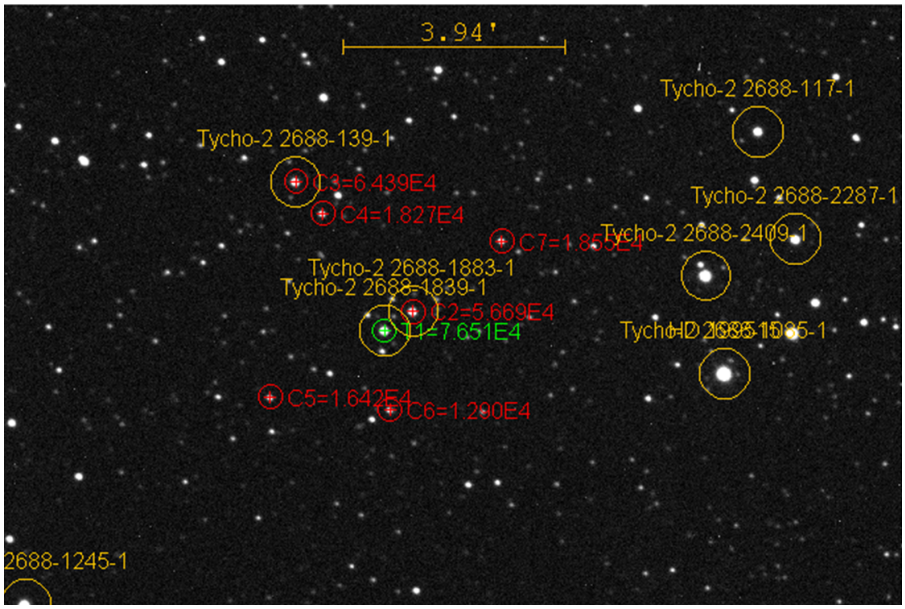


Figure 17: KELT-16b (T1) and comparison stars (C2 to C7)

Host Star		Planet	
Magnitude	11.7	Period	0.97 days
Distance from Earth	444.59 pc	Semi-major Axis	0.02044
Temperature	6430 K	Mass	2.75 M_{Jup}
Mass	1.3 $M_{\odot}$	Radius (Expected)	1.42 R_{Jup}
Radius	1.41 $R_{\odot}$	Radius (Experimental)	1.48 R_{Jup}

Table 24: KELT-16b Stellar and Planetary Parameters.

Transit			
Date	7/12-7/13	Depth (Expected)	11.5 ppt
Duration	2:29 hrs	Depth (Experimental)	12.10 ppt
Duration of observation run	10.5 hrs	T_C (UTC) (Expected)	5:15
Transit time (Eastern Time)	00:00--01:15--2:30	T_C (UTC) (Experimental)	5:12

Table 25: KELT-16b transit information.

Science Images	Flatfields

Exposure time	30 seconds	Exposure time (Morning)	1 second(s)
Temperature	-5 °C	Exposure time (Night)	1 second(s)
No. of images	480	Temperature (Morning)	-5°C
Images Deleted	Image 300	Temperature (Night)	-3°C

Table 26: KELT-16b exposure information.

The last attempt in this project for exoplanet detection is KELT-16b. It is a data set of 480; however, one image was discarded since there was a bad pixel near the target star. Telescope had issues with guiding.

Some comparisons between expected and experimental values are the following. The data provided a transit depth of 12.10 ppt, deeper than the expected 11.5 ppt. As for the transit center, expected is 05:15 UTC while the calculated is 05:13 UTC. AstroImageJ gives a planetary radius value of 1.48 R_{Jup}. This is comparable to the expected measurement of 1.42 R_{Jup}.

5 Conclusion

It is confirmed that the Keeble telescope can detect an exoplanet transit. This was achievable from the improvements made with the hardware; additionally, it is worth mentioning that the characterization of the telescope contributed as well. Regarding the CCD's abilities, the extent of what it is capable of has been investigated and there is now a grown familiarity with its features. The many adjustments made to overcome obstacles have proven to positively affect the project overall and there is a strong likelihood that these improvements will continue.

5.1 Future Work

Some considerations for future research are to investigate some of the common issues that interfered with the project. The main offender was subpar telescope tracking, which ruined many of the images. One of the possible solutions may be to purchase an autoguider. Environmental factors, such as the summer heat, caused obstacles with the CCD's thermoelectric component. As for the data, it would be

ideal to figure out how to de-trend the light curve. Another consideration is to apply to join exoplanet groups, such as TESS, and to contribute findings from continuous or follow-up observations as a ground-based telescope.

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